

Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series

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ABSTRACT

Deep learning architectures for time series modeling overwhelmingly rely on 1D serial pointwise vectors or uniform non-overlapping patch partitions. When applied to non-stationary financial markets, these paradigms suffer from severe limitations: single-scale rigidity, token boundary clipping distortion, and vulnerability to low signal-to-noise ratio (SNR) regime shifts. To resolve these challenges, we introduce Multi-Scale Overlapping Pattern Tokenization (MSOPT)—a novel framework that reformulates scalar price series into a multi-scale 2D Scale-Time Spatial Tensor. MSOPT combines dense dilated receptive field extraction (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$) with dense translation-invariant stride ($s=1$) and thresholded 1D-SAX symbolic discretization (segment mean and slope quantization) to discover human-legible visual pattern primitives ('words'). Over a rigorous 10-year walk-forward expanding window evaluation (2016–2026) across S&P 500 (SPY), Nasdaq (QQQ), Apple (AAPL), and Treasury Bonds (TLT) post 5 bps transaction costs, MSOPT achieves a +71.1% Out-of-Sample Sharpe Ratio lift over standard fixed-window baselines (0.8668 vs 0.5065 on SPY) while reducing Maximum Drawdown by 50% (from -41.91% to -22.09%).

1. Introduction

Financial time series analysis remains one of the most challenging frontiers in quantitative machine learning due to extreme non-stationarity, low signal-to-noise ratio (SNR), and non-linear regime shifts. Traditional deep sequence architectures—including Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and vanilla Transformers—process temporal data as a 1D sequence of continuous scalar points x_t in $\mathbb{R}$. This serial approach exhibits two fundamental drawbacks: first, it incurs quadratic computational complexity $O(T^2)$ over long temporal horizons; second, it fails to capture localized multi-scale visual shapes (e.g., micro-spikes, daily consolidations, macro breakouts) that human quantitative traders natively recognize.

Recent innovations in time series foundation models, such as PatchTST, attempt to mitigate sequence length by aggregating raw scalar values into contiguous uniform patches ($P=16$). However, in financial markets, uniform non-overlapping patching forces rigid partition boundaries that clip pattern inflections arbitrarily. Furthermore, non-overlapping patch boundaries lack translation invariance: shifting an input series by a single timestep completely changes token representations.

2. Related Work & Literature Gap

Time series representation learning has evolved across three dominant paradigms: (1) Pointwise Serial Models (LSTMs, N-BEATS), (2) Fixed Uniform Patching (PatchTST, ICLR 2023), and (3) Symbolic Discretization & BPE (BORF, IEEE 2024; TS-BPE, 2025). The critical research gap is that no existing framework unifies multi-scale dilated receptive field extraction, dense translation-invariant overlapping tokenization ($s=1$), 1D-SAX human-legible codebooks, and 2D scale-time spatial tensor representations within a non-stationary financial paradigm.

3. Proposed Methodology: MSOPT Framework

MSOPT reformulates time series representation through a 4-stage pipeline:

- **Multi-Scale Receptive Fields:** Dense sampling across window sizes w in $\{4,8,16,32\}$ and dilations d in $\{1,2,4\}$ with dense stride $s=1$ for 100% translation invariance.
- **1D-SAX Discretization:** Subseries are quantized into discrete mean (α_{μ}) and slope (α_{β}) symbols, creating a robust noise-filtering vocabulary.
- **2D Scale-Time Spatial Grid:** Tokens are indexed into a 2D matrix H in $\mathbb{Z}^{N_{\text{scales}} \times T}$.
- **PyTorch Conv-Transformer:** 2D spatial convolution Inception blocks capture inter-scale pattern composition before Multi-Head Self-Attention.

4. 10-Year Out-of-Sample Walk-Forward Results (2016–2026)

Asset	Model	Sharpe Ratio	Sortino Ratio	Max Drawdown
SPY	Baseline (Fixed Windows)	0.5065	0.5979	-41.91%
SPY	MSOPT Tokens (Ours)	0.8668	1.0847	-22.09%
QQQ	Baseline (Fixed Windows)	0.5731	0.7049	-46.82%
QQQ	MSOPT Tokens (Ours)	0.8709	1.0536	-41.57%
AAPL	Baseline (Fixed Windows)	0.0554	0.0678	-72.41%
AAPL	MSOPT Tokens (Ours)	0.3745	0.4999	-38.87%
TLT	Baseline (Fixed Windows)	-2.1339	-2.8557	-95.09%
TLT	MSOPT Tokens (Ours)	-0.0675	-0.0995	-29.43%

5. Conclusion & Future Directions

MSOPT establishes a new paradigm in time series pattern recognition. By combining multi-scale 1D-SAX tokenization with translation-invariant overlapping stride ($s=1$) and 2D spatial Conv-Transformers, MSOPT delivers substantial Sharpe ratio gains (+71.1% on SPY) and robust regime crash protection post 5 bps transaction costs.